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UNIVERSITY OF SOUTH CAROLINA

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CSCE 190: Computing in the Modern World

Class 7 – Problem Validation (Solving with Computing)

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Carolinian Creed: "I will practice personal and academic integrity."

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Outline

- Introduction Section
  - Recap from Last Class
- Main Section
  - Problem Solving Process
  - Validating the Problem
  - Selecting Problems that Matter
- Conclusion Section
  - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 6

We talked about

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Assignment 6

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see ***problems that matter***
- Write a problem statement
  - The group will have one problem statement
  - Add your statement to your GitHub in a sub-folder called ‘assignment-6’
    - Do mention all the students in your group

Computing in the News

- Impact of AI Slop on Research papers, Oct 2026, https://startupfortune.com/arxiv-now-limits-every-researcher-to-two-paper-submissions-a-month/
  - Arxiv is a site for Research pre-prints (pre-publication papers). Seeing drastic rise in submissions, esp. cs.AI which has 1/3 categorized by authors. Instructor is one of 10 moderators across the world.
  - "Fair Moderation, Equitable Access, and AI: arXiv's Updated Rate Limit Policy," – announcement blog
  - In May 2026, a one-strike rule: any author caught submitting a paper with hallucinated references or leftover LLM instructions baked into the text faces a one-year ban, followed by a requirement that all their future submissions clear peer review before they can even be posted as a preprint.

  [No one wants to deal with AI slop]

- The Economics of Using AI to Churn Out Code Are Looking Worse Than Ever, May 2026, https://futurism.com/artificial-intelligence/economics-ai-churn-code

  [Code generation is just the start!]

Computing in the News

- Anthropic meetings with religious leaders, Oct 2026, https://www.nytimes.com/2026/09/29/us/anthropic-claude-morals-ai.html; https://the-decoder.com/anthropic-co-founder-reportedly-told-religious-leaders-he-fears-having-created-something-that-suffers-perpetually/
  - Anthropic trying to convince representatives from various religions that their system's behaviors can be acceptable (in the long run)

  [More marketing ?]

- Alibaba open-sources medical AI model that can detect cancer and nearly 150 conditions, Oct 2026, news - https://sc.mp/z3bsy; paper - https://www.science.org/doi/10.1126/science.aec6129
  - Open-sourced an artificial intelligence model capable of identifying nearly 150 abdominal conditions – including cancers – by reading computed tomography (CT) scans
  - Training data is key

  [Applications that matter]

Main Section

Problem Solving with Computing

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Problem Solving Process

Receiving the problem ***{from client}***

- Client creates **requirements** document

- **Understanding the problem and scoping**
  - Developer creates **specification**

- **Solving**
  - Understand and implementing with tools for solving the problem: **Design docs, test plans**
  - Create solutions: **Code, Release**
  - Evaluate solutions: **Test logs**

- **Delivering *{to customers}***

A generic process involves:

- Customer:
  - requirements
- Developer:
  - specification
  - design
  - development (coding)
  - testing
  - release
- Customer:
  - evaluation
- acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

Validating the Problem

- **Objective**: Understanding the problem and scoping
- Think and articulate – what will be
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software, solution, documentation
  - Criteria for problem being solved: e.g., metrics

  - Considerations
    - Current practices/ processes
    - Why would someone use your solution to the problem?
    - **If you can't identify inputs, outputs, and evaluation metrics, the problem is most likely ill-defined computationally. Needs more work!**

SMART Metrics

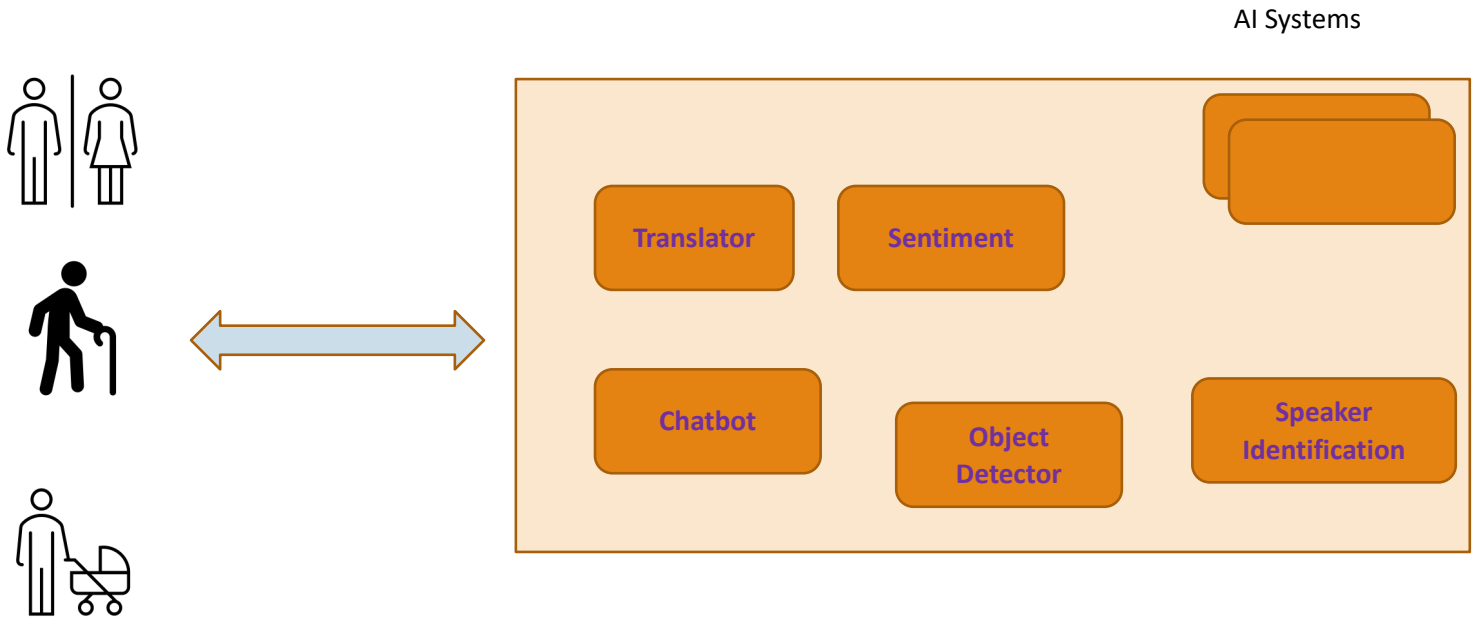
- ***Specific:*** Targeting a particular area for improvement
- ***Measurable:*** Quantifying, or at least suggesting, an indicator of progress
- ***Assignable:*** Defining responsibility clearly
- ***Realistic:*** Outlining attainable results with available resources
- ***Time-related:*** Including a timeline for expected results

Source: https://en.wikipedia.org/wiki/SMART_criteria

Selecting Problems that Matter

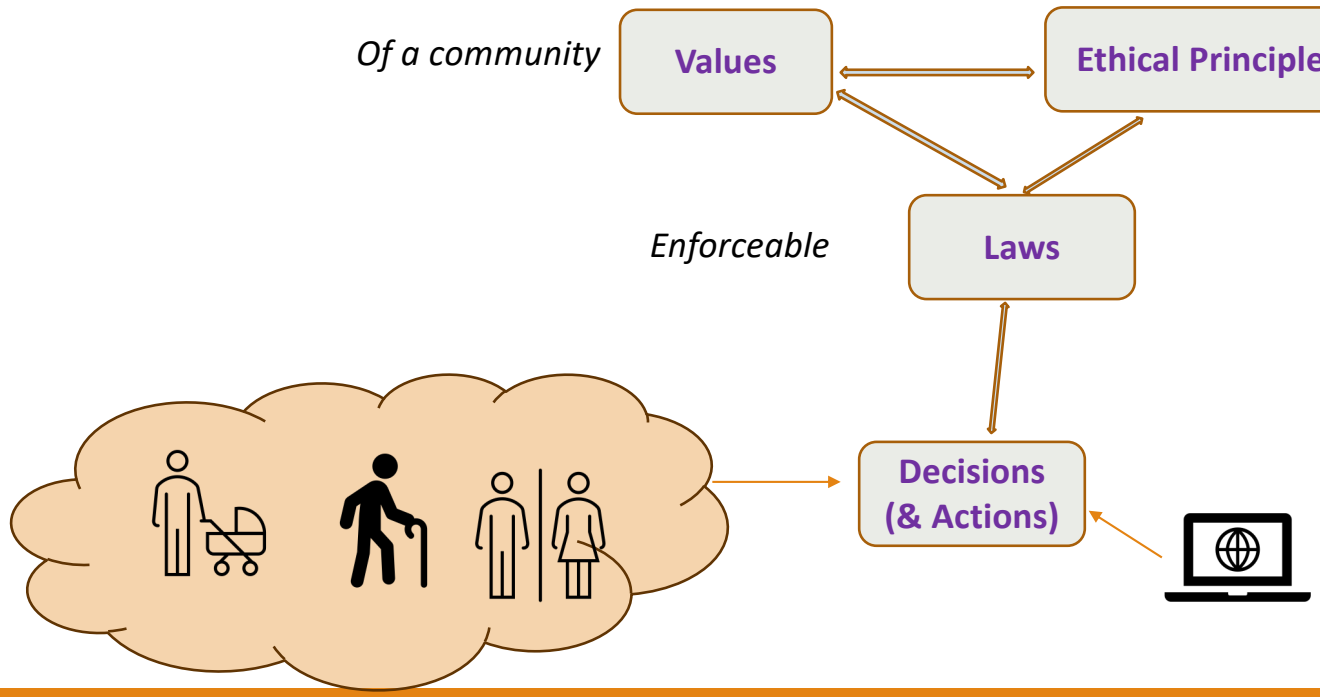
- Answer
  - For whom is the problem important? **[A lot of people]**
  - What happens if you solve it well? **[With change the status quo for the better]**
  - Do you have the inputs to get started? **[Yes? Otherwise, re-think problem or inputs]**
  - Do you have the resources/ are you best qualified to solve the problem? Do you need to augment the team? **[Yes? Otherwise, pass or get help]**

Technology and People



Trust: *Can people trust* ***new system*** *to perform capably, consistently, and with human values?*

Values, Ethics, Law, Decisions (Actions) with Trust



Trust: *how can, humans or machines (AI) or in collaboration together, take decisions and actions following the same **values**, **ethical** principles and **laws ?***

Assumption: *they should follow same values, principles and laws.*

Problem Example 1: Student Meals Reco

- **Problem**: build a system to help students choose meals during study days
- Validating the problem:
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software (app)
  - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
  - Recommendation methods
- Trustworthiness ecosystem
  - Values, ethical principles, laws
  - Users impacted

Problem statement criteria

- Are focused on the user (avoid using “we” or “the product”)
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Problem Example 2: Noble Prizes

- **Problem**: build a system to contextualize about Noble Prizes wins
- Validating the problem:
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software (app)
  - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
  - **Identifying topics and trends**
- Trustworthiness ecosystem
  - Values, ethical principles, laws -
  - Users impacted – ***students making future decisions based on trends ?***

Problem statement criteria

- Are focused on the user (avoid using "we" or "the product")
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Noble Prizes: Non-functional Requirements/ Specification

- Winners background
  - Name, gender, age, country, institution
  - Should or shouldn't be given ? Any disclaimer
- What do the awards in a year tell about
  - The past ?
  - The future ?
- Themes
  - Terminology changes – how to handle them

Problem Categories Identified

1. Helping students with
    1. class commitments
        1. books
    2. meals
    3. parking
    4. finances

Suggested

1. Helping students
    1. With class commitments
    2. With meals
    3. With finances (?)
    4. …

Snapshot from Different Sections

Section 1	Section 2	Section 4
1. students on scooters (harm) 2. Students managing money in college 3. Students finding housing (off-campus) 4. Students managing assignments **5. (all) Sidewalks uneven** 6. Students managing finances – 1st year 7. Students: saving meal swipes (7!)	**1. Students: textbook overprices and requirements** 2. (All) drunk driving at 5-points **3. Students: on-campus laundry** 4. Students: USC bus trackers are unreliable 5. Students: non-reserved parking 6. Students: double-deck parking	1. Students: pedestrian safety, accessibility 2. Students: campus parking and transportation 3. {} – problem not noted 4. Students: college tuition payment **5. Students: time management and resources**

Class Activity

- If you have a group, refine problem
- If you don’t, form or join a group. If new, come up with a problem
- Prepare for Assignment-7

Assignment 7: Same for All in a Team

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see ***problems that matter***
- Write a problem statement
  - The group will have one problem statement
  - Add your statement to your GitHub in a sub-folder called 'assignment-6'
    - Do mention all the students in your group

- Write for **client**
  - **Requirements**: what will meet the need?
- Write for **developers**
  - **Specification**: what will the system-to-be built do
  - **Test plan**: how will we show that the built system works to meet the specification
- Add requirements, specification, and test plan to your Github in a sub-folder called 'assignment-7'
  - Do mention all the students in your group

See sample documents at: https://github.com/biplav-s/course-cmw-f26/tree/main/software-dev

Concluding Section

Conclusion

We talked about

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics) + Lab	Github account, repo
4	HTML basics, IDE (VS Code)	None
5	More HTML, CSS, Problem Solving	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	AI – LLMs/FMs, Generative AI, Issues and Challenges	Problem Solving with GenAI
10	Building a resume, LinkedIn	Draft resume in PDF
11	Review resumes	Review resumes, update Github
12	Convert resume to HTML	Begin conversion, update Github
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	**All extensions end**

Ask Me Anything